

The human genome sequence was finished in.... 2021?



The Telomere-to-Telomere (T2T) Consortium has completed a challenging 8% of the human genome left unresolved by the initial Human Genome Project. In this data visualization, each chromosome begins at bottom right and wraps around, with chromosomes X and 1 through 22 arranged from the outside in (chromosome Y is not shown). The newly completed regions are highlighted in red.

Podcast:

[Scientists Release The First Fully Complete Human Genome \(sciencefriday.com\)](https://www.sciencefriday.com)

Reminder: Self-Assessment Exercise on Wednesday March 12

- Same format as previous SAEs
- Multiple choice, True/False
- Goals: reflect, collaborate, assess
- 8:35 am: lecture wrap-up
- 8:55 am: SAE

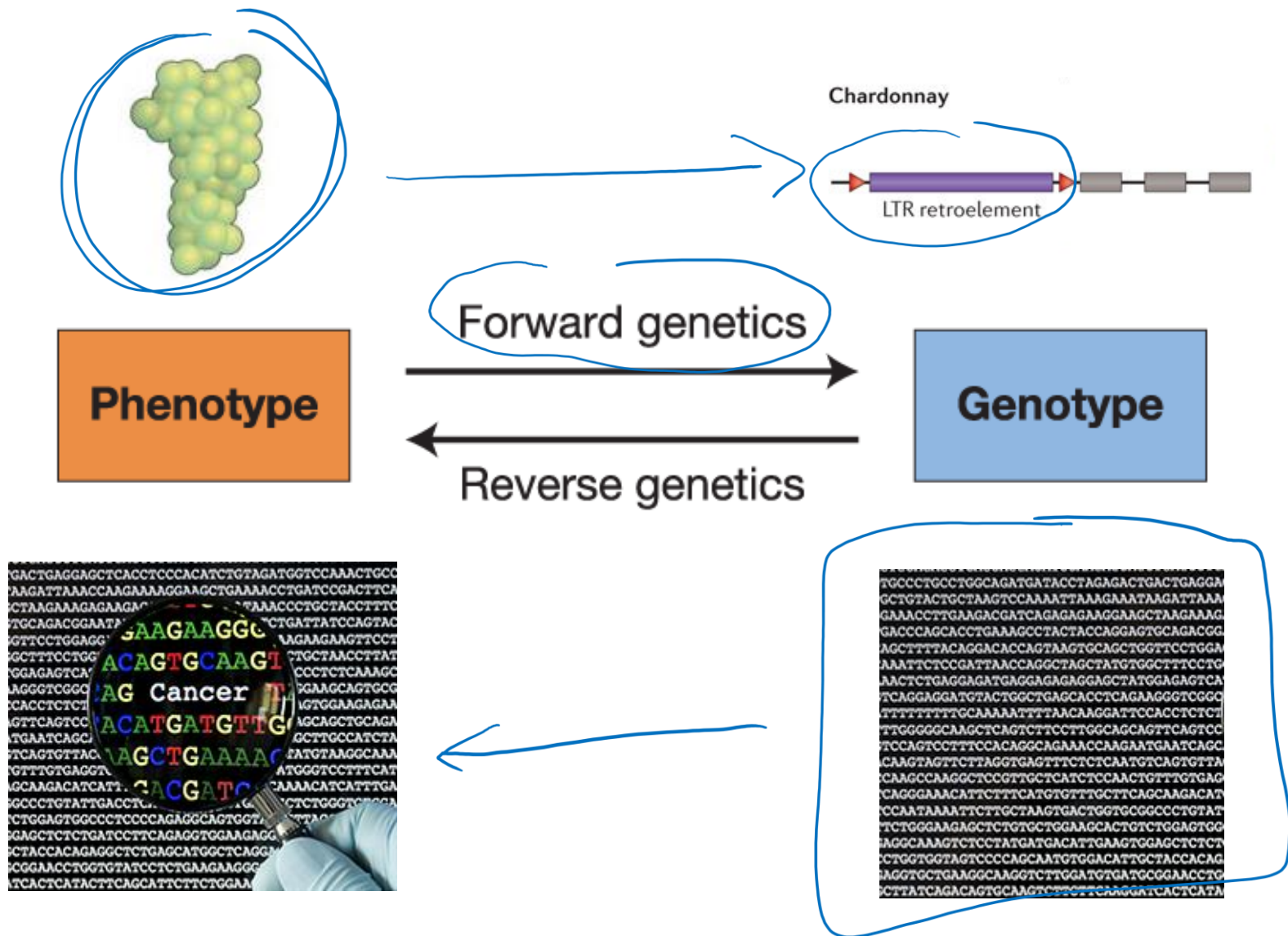
Bioinformatics and Genomics

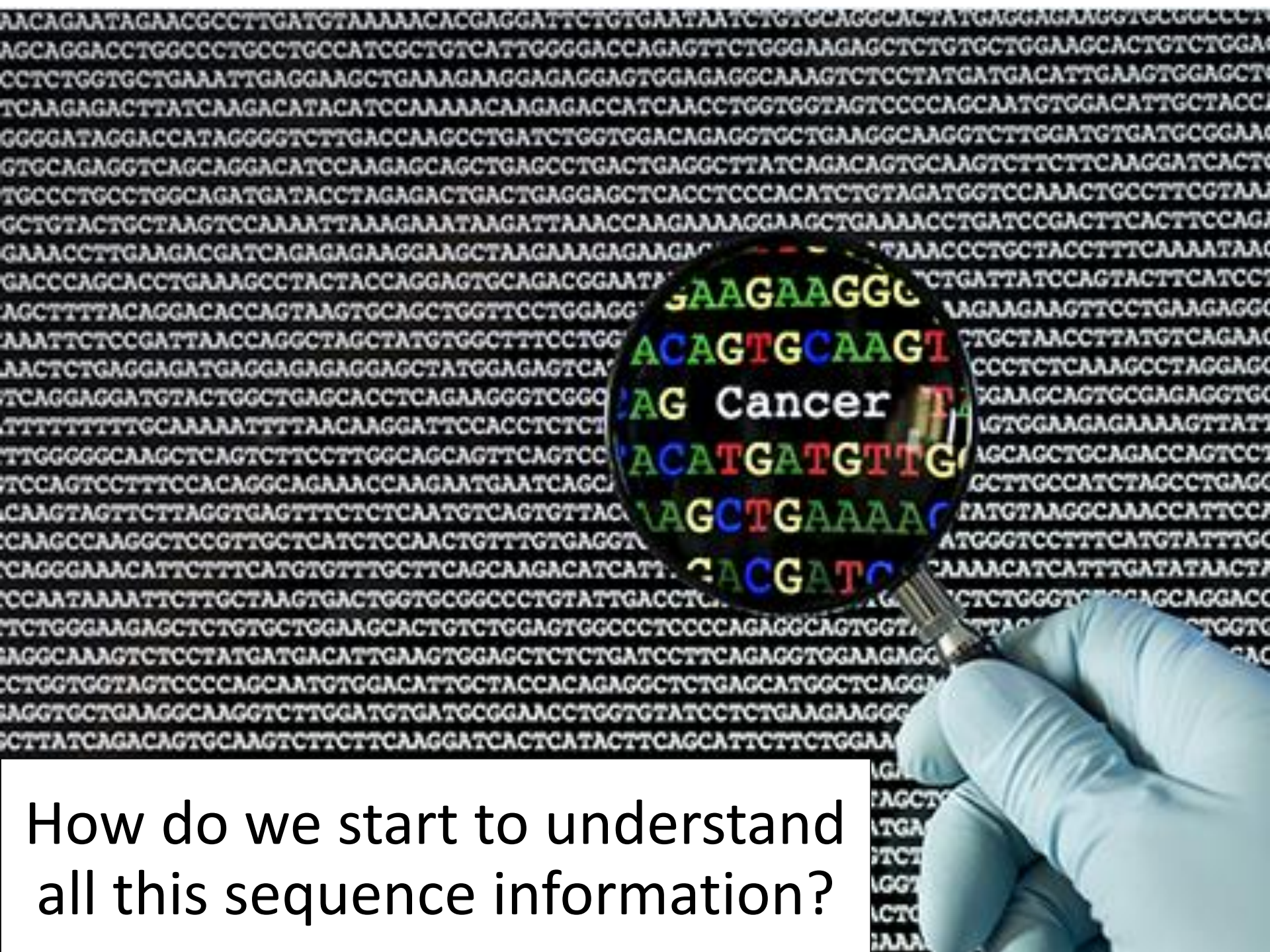
Bioinformatics and genome annotation

Comparative genomics

Functional genomics and reverse
genetics

How can we read and understand a genome?

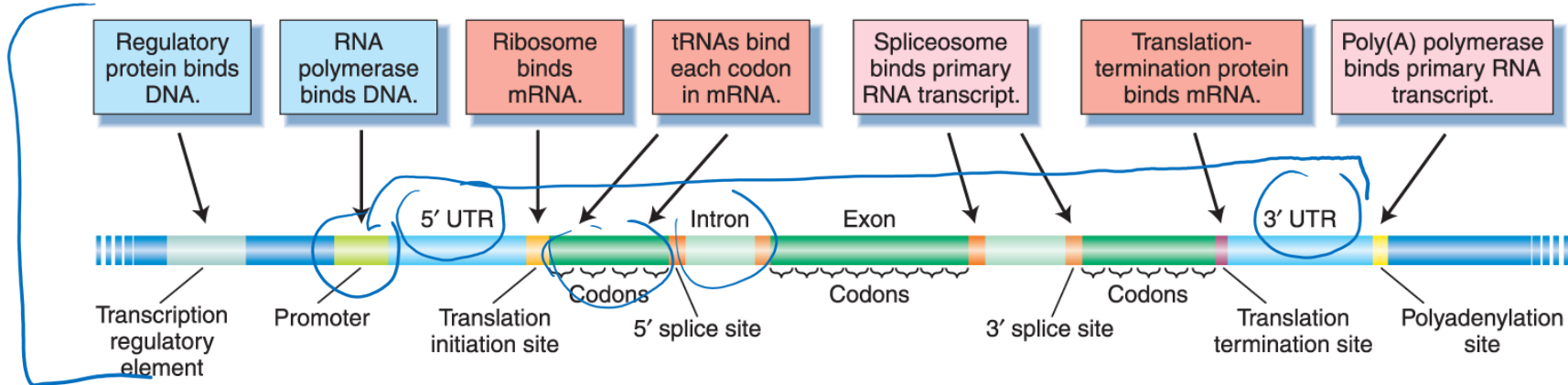




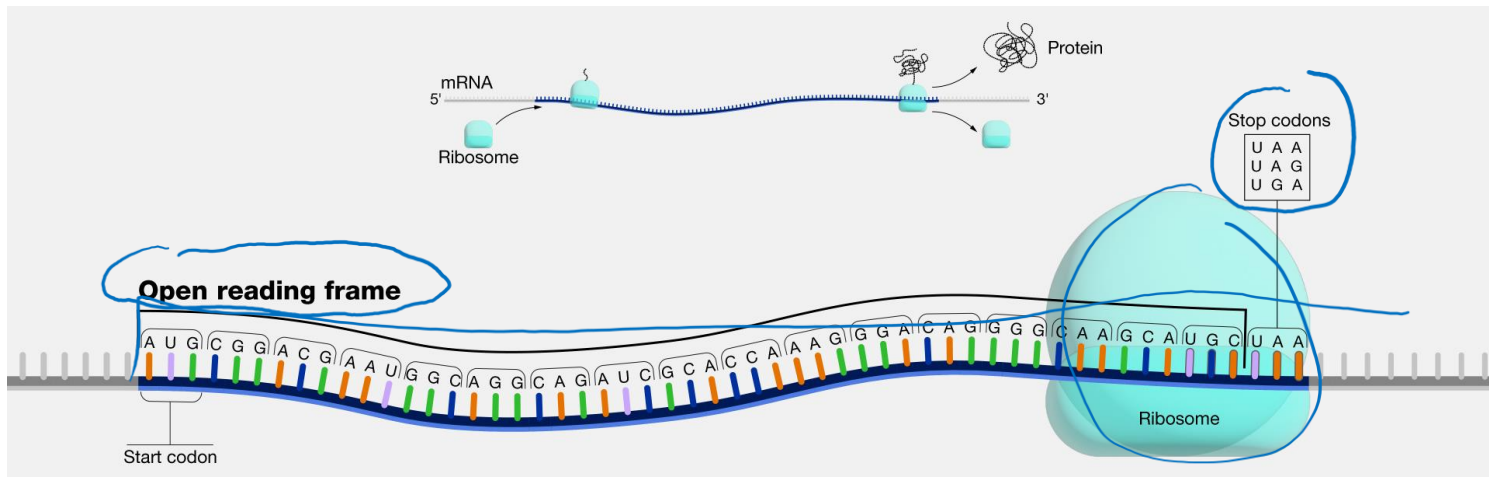
How do we start to understand all this sequence information?

Start with what we know best

Look for sequences associated with protein-encoding genes

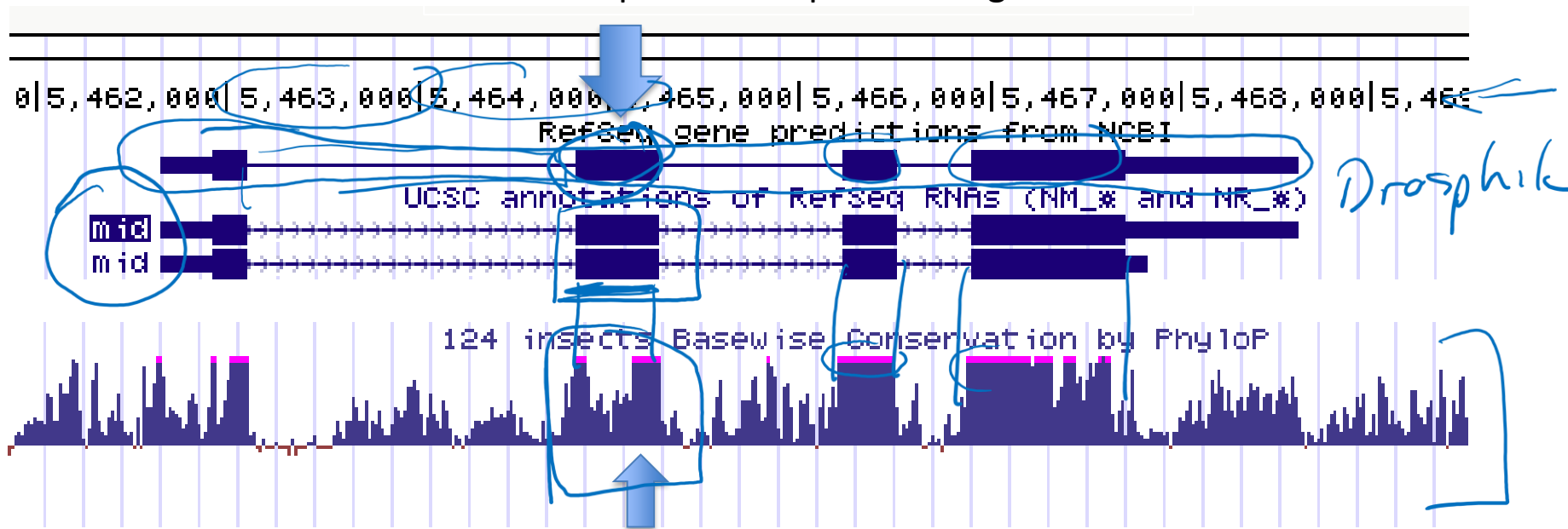


First step: look for open-reading frames (“ORFs,” i.e. stretches of codons that aren’t stop codons) as evidence of possible exons/genes.



Check for conservation of a predicted ORF across species as evidence that it is part of a real gene.

Bars show predicted open reading frames

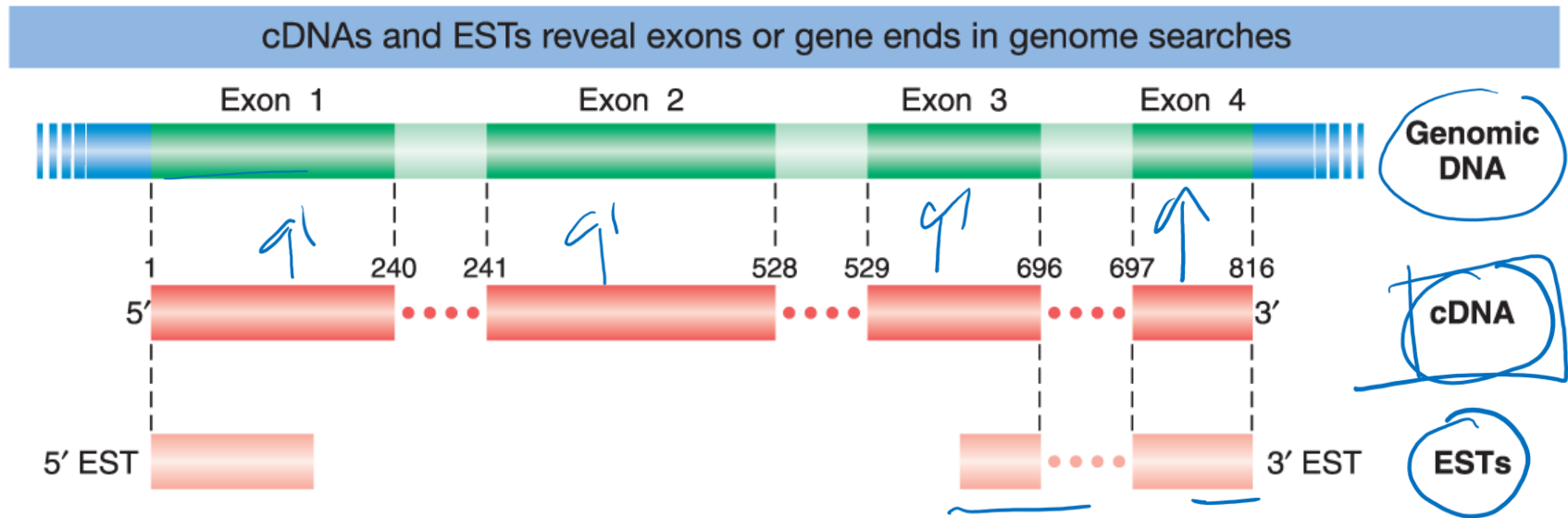


Peaks show levels of sequence conservation across 100+ insect species

peak conservation aligns with ORFs → suggests that they are actual exons of a real gene

Look for cDNAs that map to the putative gene

Identification of a cDNA that maps to a predicted ORF provides the best evidence that the ORF is part of a real gene.



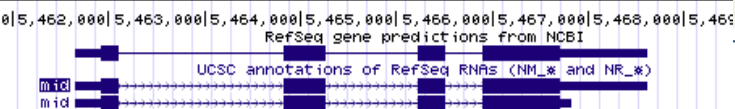
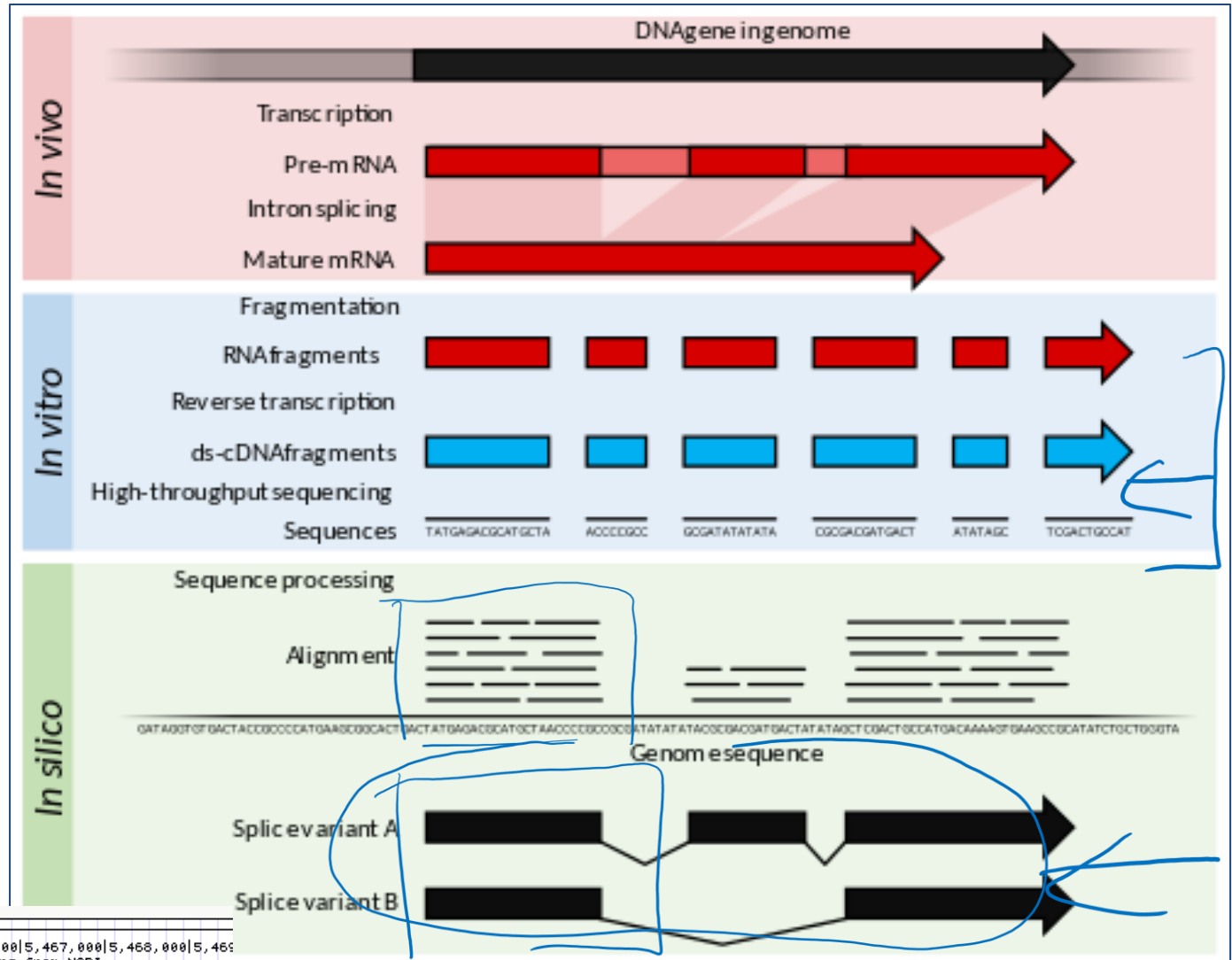
An EST ("expressed sequence tag") is a cDNA that has not been completely sequenced.

Look for cDNAs that map to the putative gene

Cells transcribe genes to make mRNA.

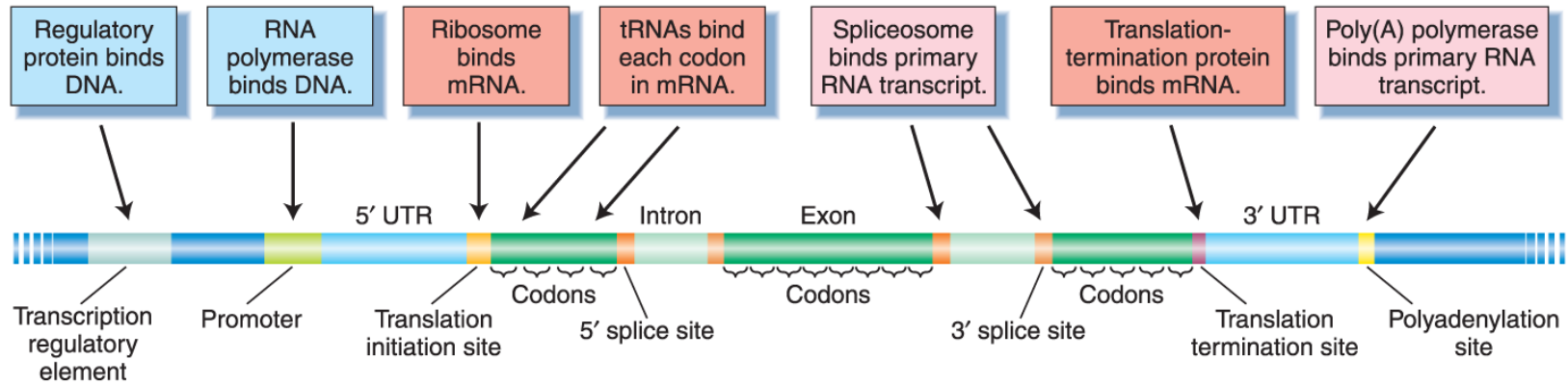
mRNA is isolated, reverse transcription generates cDNA copies of mRNAs, cDNAs are sequenced.

Sequences are aligned to the genome sequence to determine where the mRNA came from.



Other sequence features associated with genes

Protein binding sites



Codon bias

Codon	Human	Drosophila	E. coli
Arginine:			
AGA	22 %	10 %	1 %
AGG	23 %	6 %	1 %
CGA	10 %	8 %	4 %
CGC	22 %	49 %	39 %
CGG	14 %	9 %	4 %
CGU	9 %	18 %	49 %
Total number of arginine codons	2403	506	149
Total number of genes	195	46	149

Species-appropriate “codon bias” can provide evidence that an ORF is part of a real gene.

Genetic code: same for all organisms

Amino Acids			Codons					
Alanine	Ala	A	GCA	GCC	GCG	GCU		
Cysteine	Cys	C	UGC	UGU				
Aspartic acid	Asp	D	GAC	GAU	GAC	GAU		
Glutamic acid	Glu	E	GAA	GAG				
Phenylalanine	Phe	F	UUC	UUU				
Glycine	Gly	G	GGA	GCG	GGG	GGU		
Histidine	His	H	CAC	CAU				
Isoleucine	Ile	I	AUA	AUC	AUU			
Lysine	Lys	K	AAA	AAG				
Leucine	Leu	L	UUA	UUG	CUA	CUC	CUG	GUU
Methionine	Met	M	AUG					
Asparagine	Asn	N	AAC	AAU				
Proline	Pro	P	CCA	CCC	CCG	CCU		
Glutamine	Gln	Q	CAA	CAG				
Arginine	Arg	R	AGA	AGG	CGA	CGC	CGG	CGU
Serine	Ser	S	AGC	AGU	UCA	UCC	UCG	UCU
Threonine	Thr	T	ACA	ACC	ACG	ACU		
Valine	Val	V	GUA	GUC	GUG	GUU		
Tryptophan	Trp	W	UGG					
Tyrosine	Tyr	Y	UAC	UAU				

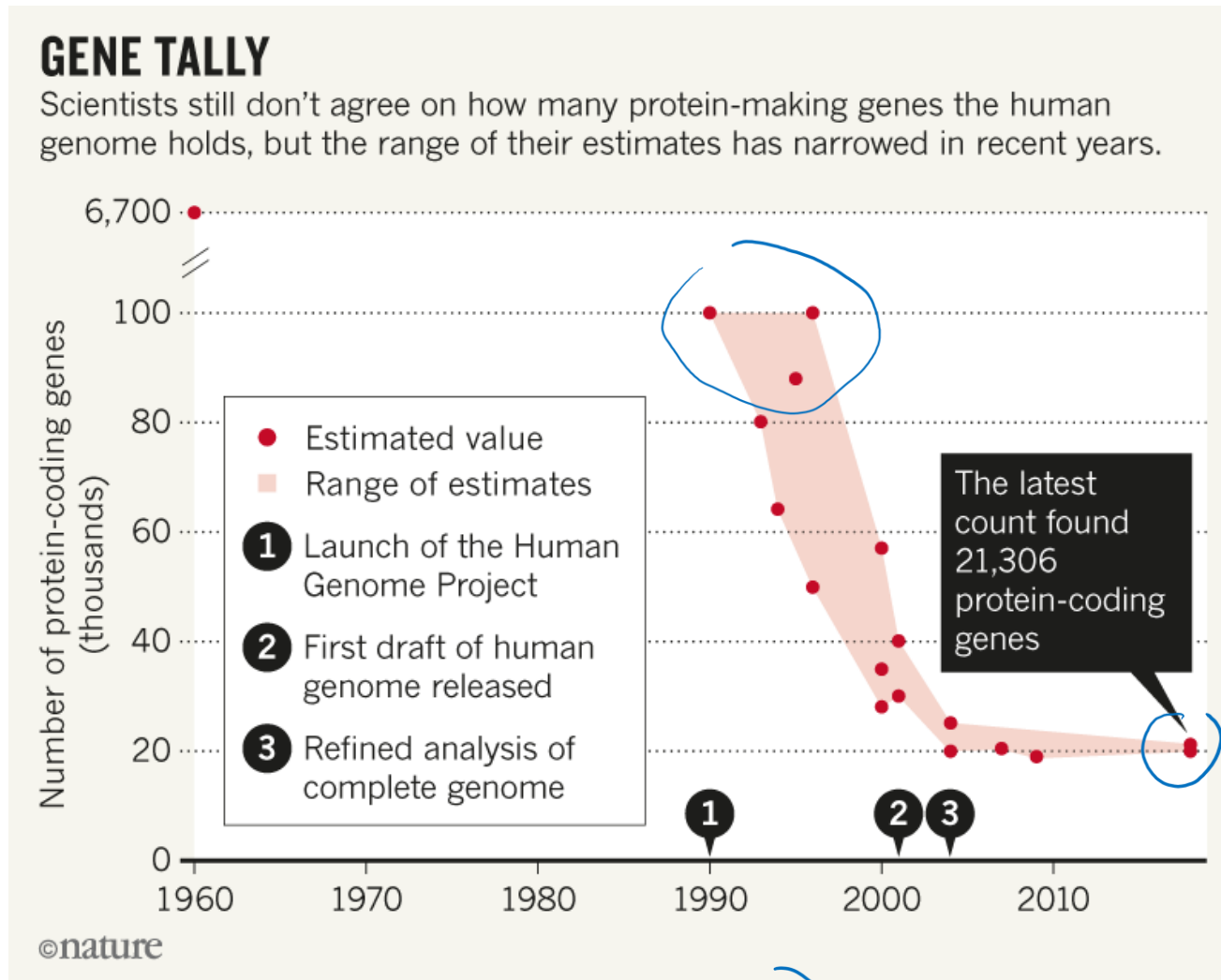
Codon usage: can differ across species

Codon	Human	Drosophila	E. coli
Arginine:			
AGA	22 %	10 %	1 %
AGG	23 %	6 %	1 %
CGA	10 %	8 %	4 %
CGC	22 %	49 %	39 %
CGG	14 %	9 %	4 %
CGU	9 %	18 %	49 %
Total number of arginine codons	2403	506	149
Total number of genes	195	46	149

Most amino acids can be coded for by more than one codon, but different species have different frequencies of synonymous codon usage. This codon preference reflects relative tRNA abundance and can be a signature of an ORF that's part of a real gene.

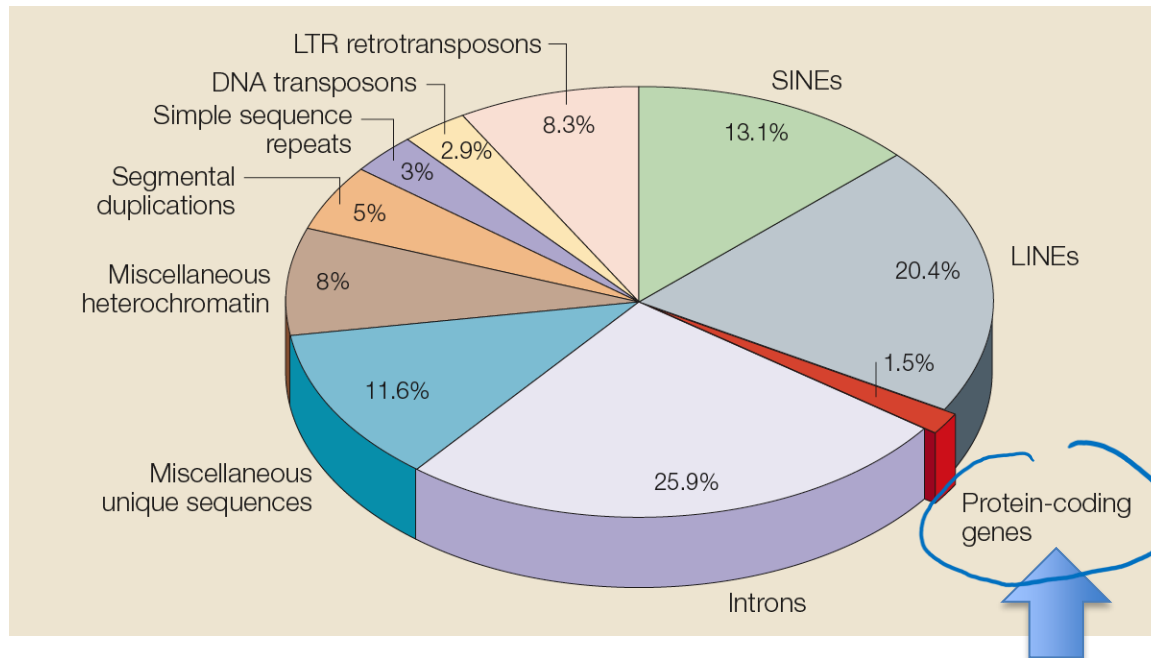
The structure of the human genome

After all this analysis, how many genes are there?



The structure of the human genome

Protein-encoding genes are rare. Repeated sequences are common.

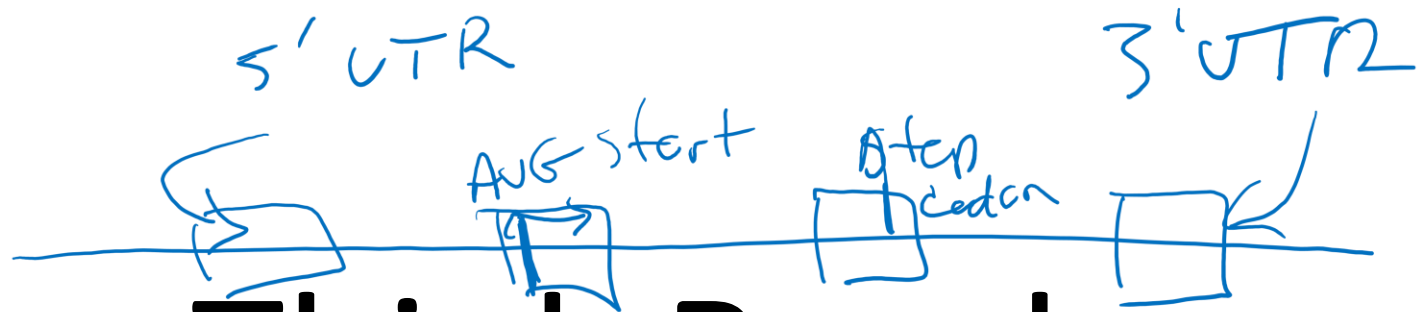


~3% of the human genome is made up of exons of protein-encoding genes.
(exons plus introns plus regulatory sequences: = ~28%)

~1% encodes protein sequences.

~45% is made up of repetitive sequences.

- exons not in open frames



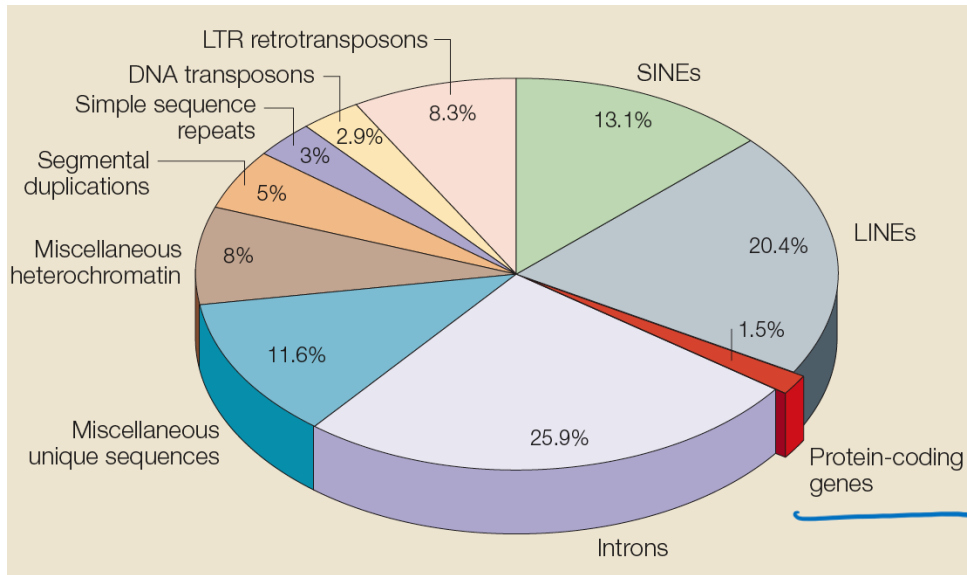
Think Break



→ non coding RNA → yes, maybe not ok
- enhancers → ~~is~~ in genome, but not ^{*} in exons

The structure of the human genome

Beyond protein-encoding genes, what is the rest of the genome doing?



Pseudogenes

Functional RNA transcripts

- microRNAs
- siRNAs
- piRNAs
- long non-coding RNAs

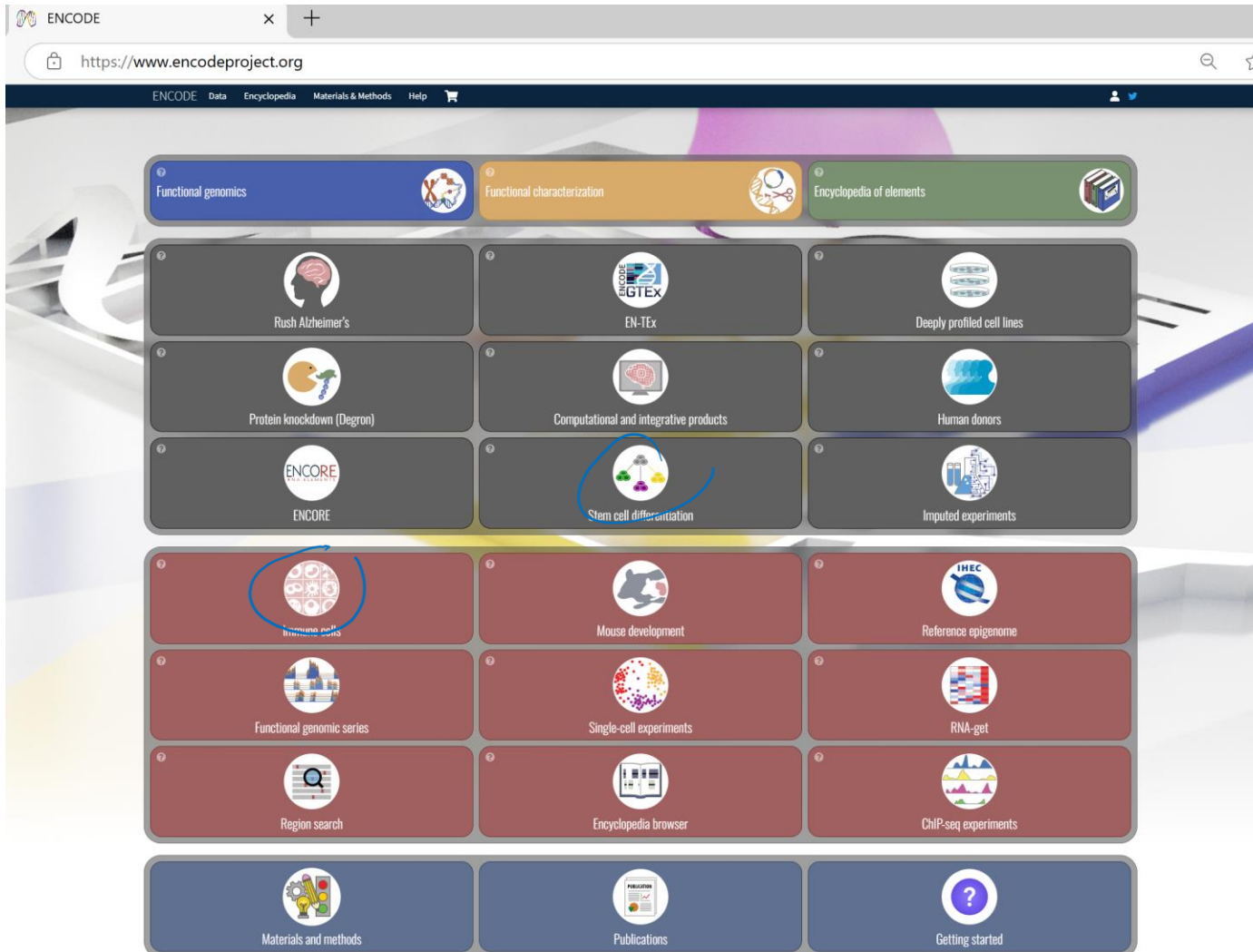
Gene regulatory elements

- enhancers
- transcription factor binding sites

How can we figure this out?

ENCODE: Encyclopedia of DNA elements

goal: to identify all functional elements within the human genome



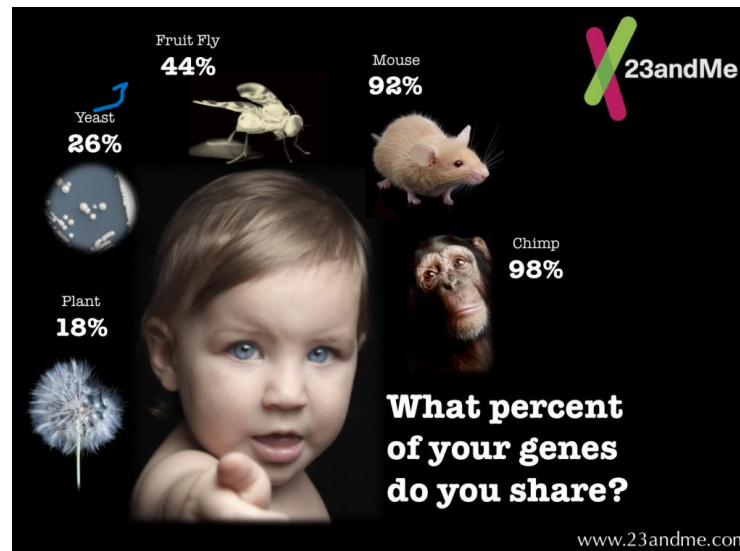
All transcribed regions (~80% of the genome!)

All regions that control transcription

Tools for functional analysis (e.g. knockdowns)

sequences Comparative Genomics

Are genes conserved between species? What information beyond coding sequences is encoded in the genome? Comparison of genomes reveals evolutionary relationships between organisms and identifies both conserved and unique elements.



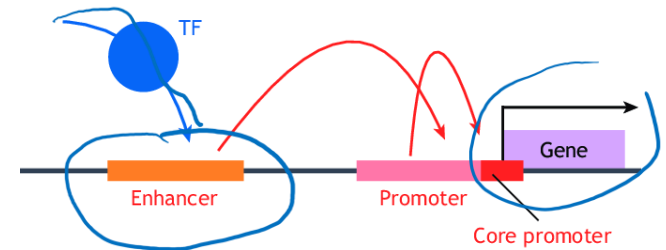
- Comparison **between organisms**: to compare types of genes and their arrangement in the genome, to infer information about genome structure and evolutionary processes.
- Comparison **within an organism**: to identify gene families and gene duplication.
- Comparison **between individuals of the same species**: to identify differences associated with phenotypes or disease.

Sequence comparisons between species

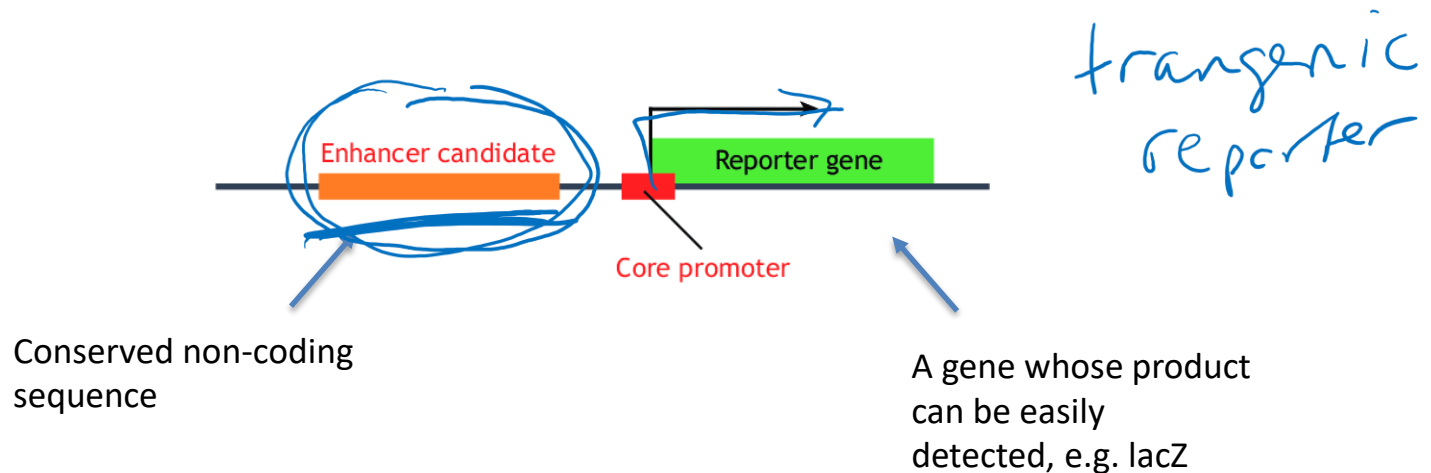
Sequence conservation suggests functional importance

Comparing genomes between species can identify important regulatory regions.

Hypothesis: conserved non-coding sequences might be regulatory elements that control gene expression



Experiment to test hypothesis: can the conserved sequence drive gene expression?



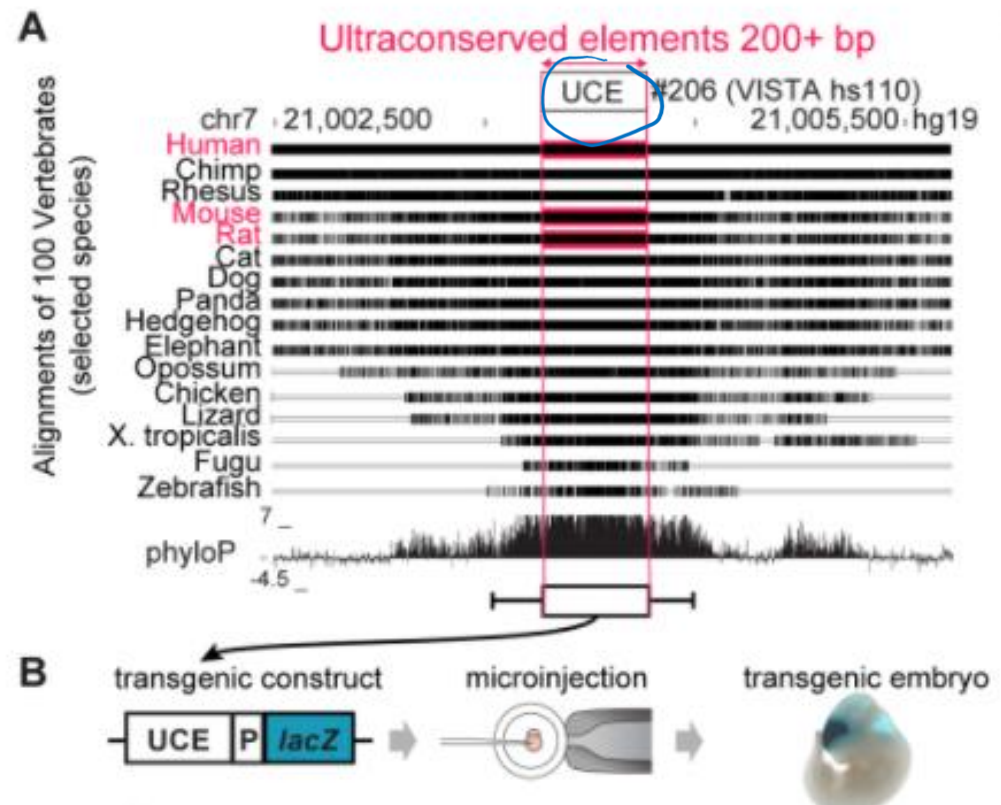
Conservation of non-coding sequences can identify functional elements

Comparison of human, mouse and rat genomes has found more than 5000 sequences of >100bp and 481 sequences of <200bp that are absolutely conserved.

Many are conserved in other species.

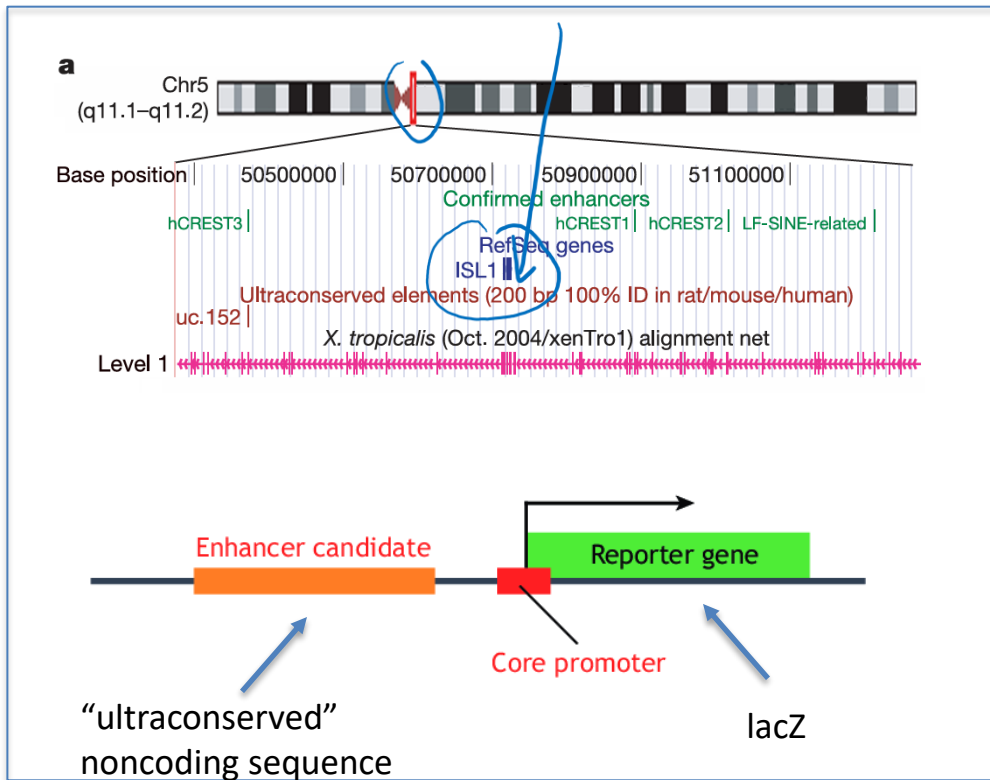
Many are found in non-coding regions and seem to regulate genes involved in development.

Study these using transgenic reporters.



Conservation of non-coding sequences can identify functional elements

ISL1



Rationale: Sequences that are important will be conserved over evolutionary time. This notion applies to both coding and non-coding sequences.

Hypothesis: This conserved non-coding region may play a role in gene regulation.

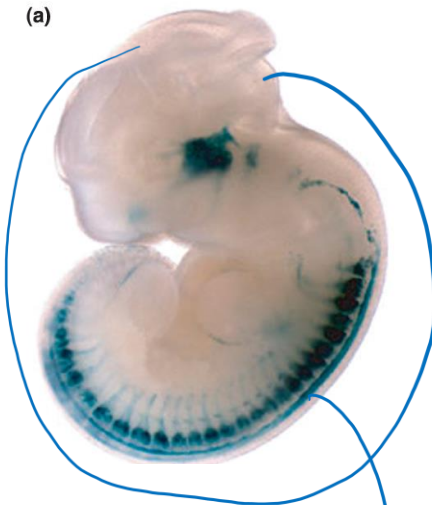
Prediction: If so, the human version should be able to function in the mouse.

Experimental approach: generate a "transgenic reporter" in which this region is placed upstream of a reporter gene whose expression is easy to detect.

Conservation of non-coding sequences can identify functional elements

Testing the role of a conserved element in gene regulation

(a)



lacZ expression driven by ultraconserved sequences from the human ISL1 locus

(b)



endogenous expression of the mouse *ISL1* gene

blue = lacZ expression

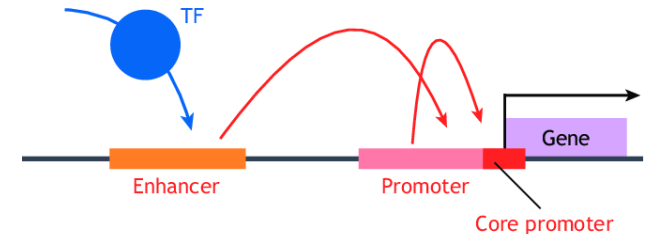
Observation: Ultraconserved non-coding sequences from the human *ISL1* gene can recapitulate the expression of the mouse *ISL1* gene during embryogenesis.

Interpretation: This ultraconserved element is a regulatory element controlling expression of the *ISL1* gene in both mouse and humans.

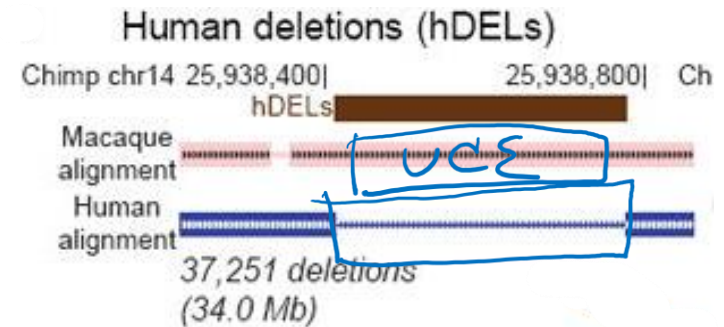
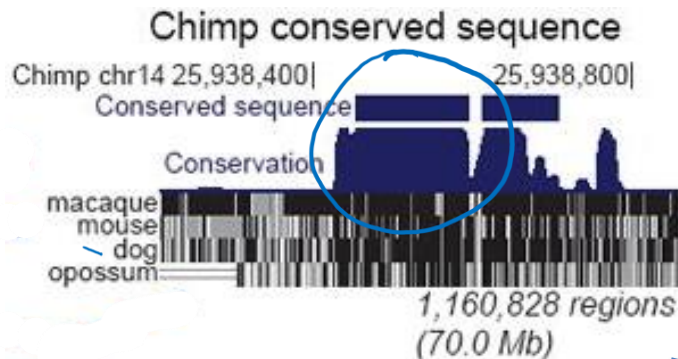
- (a) A mouse embryo bearing a transgenic reporter consisting of ultraconserved element near the human *ISL1* gene coupled to a reporter gene. and injected into fertilized mouse oocytes. The regions where the gene is expressed are stained dark blue or black. The reporter gene is expressed in the head and spinal cord of a transgenic mouse, as seen here on day 11.5 of gestation.
- (b) The native pattern of expression of the mouse *ISL1* gene on day 11.5 of gestation corresponds closely to that of the reporter.

Comparing genomes between species can identify important regulatory regions.

Hypothesis: conserved non-coding sequences might be regulatory elements that control gene expression



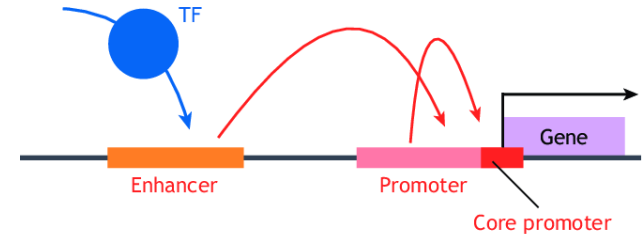
Approach: Look for blocks of DNA sequence that are conserved between chimps and macaques but not found in the human genome. Might loss of these regions explain some of the differences between chimps and humans?



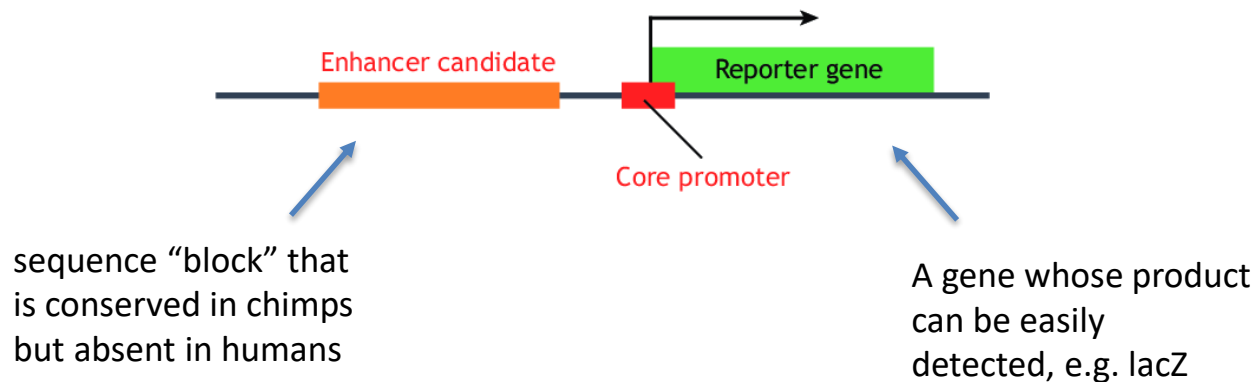
Human deletions that contain
chimp conserved sequence (hCONDELs)
583 deletions

Comparing genomes between species can identify important regulatory regions.

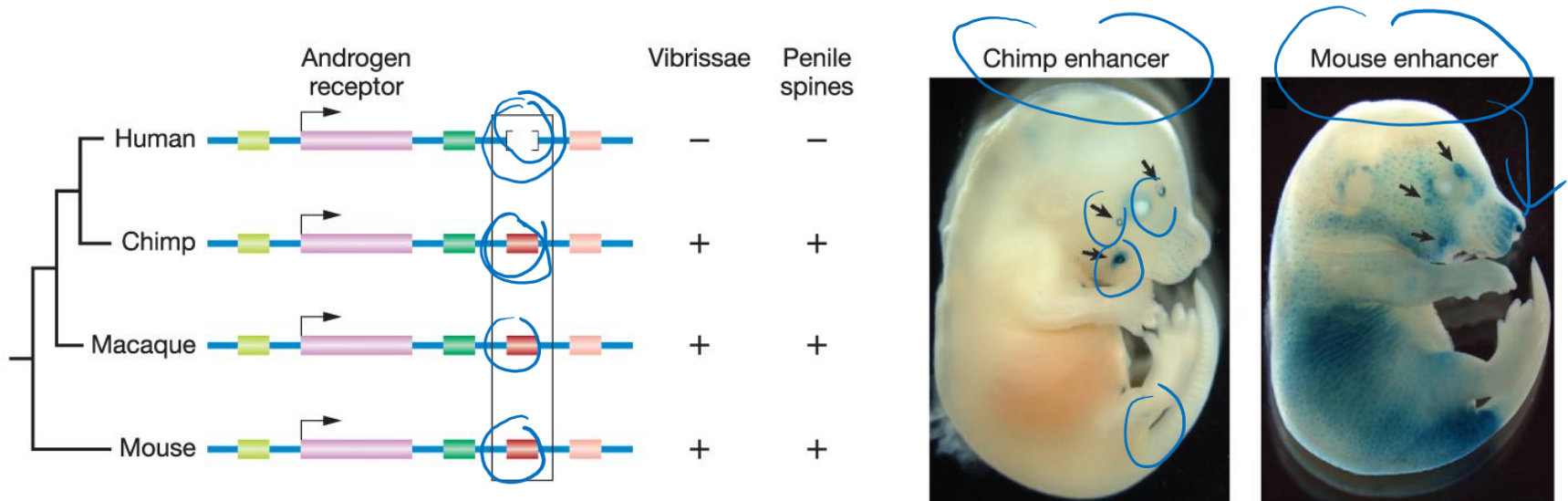
Hypothesis: conserved non-coding sequences might be regulatory elements that control gene expression



Experiment to test hypothesis: can the conserved sequence drive gene expression in the mouse?



Testing the role of the conserved enhancer that has been deleted in humans



Result: This enhancer can drive expression in two different structures that are present in mammals like mice and chimps but have been lost in humans.

Interpretation: Loss of an ancestral vibrissae/penile spine enhancer in humans is correlated with corresponding loss of sensory vibrissae and penile spines.

Sequence comparisons between species

Genome organization, synteny

Comparison of genomes between species shows striking similarities in genome organization.



99% of mouse genes have a homolog in the human genome



99% of human genes have a homolog in the mouse genome

Overall genome organization, including the relative order of genes and non-coding regions, is also highly conserved.

The conserved order of genes between the two genomes is called SYNTENY.

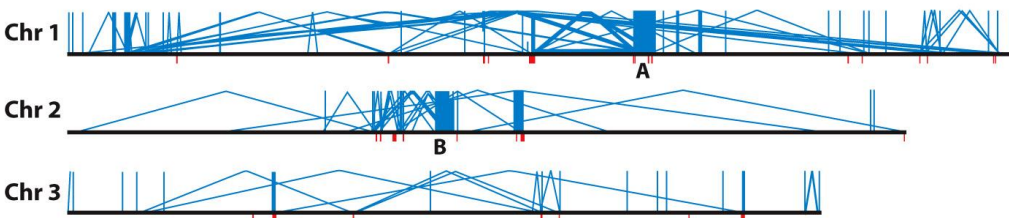
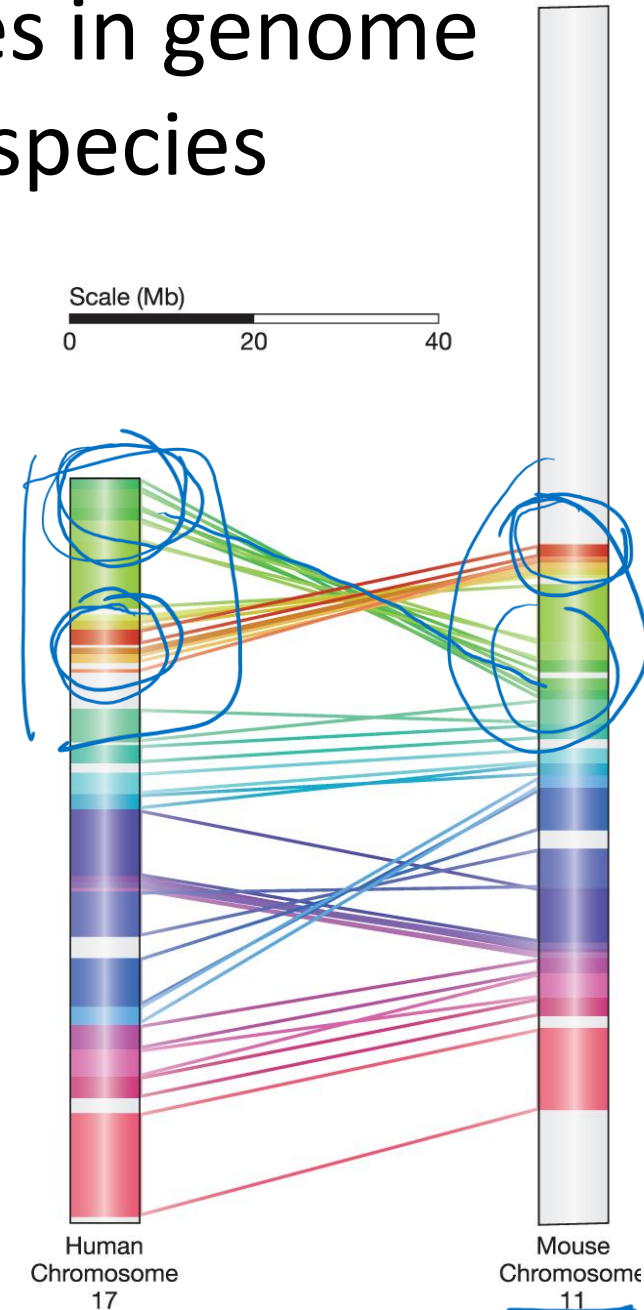
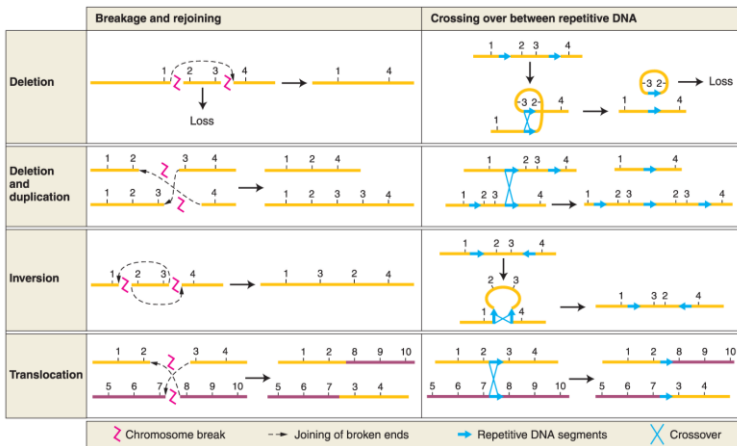
Synteny describes similarities in genome organization between species

Observation:

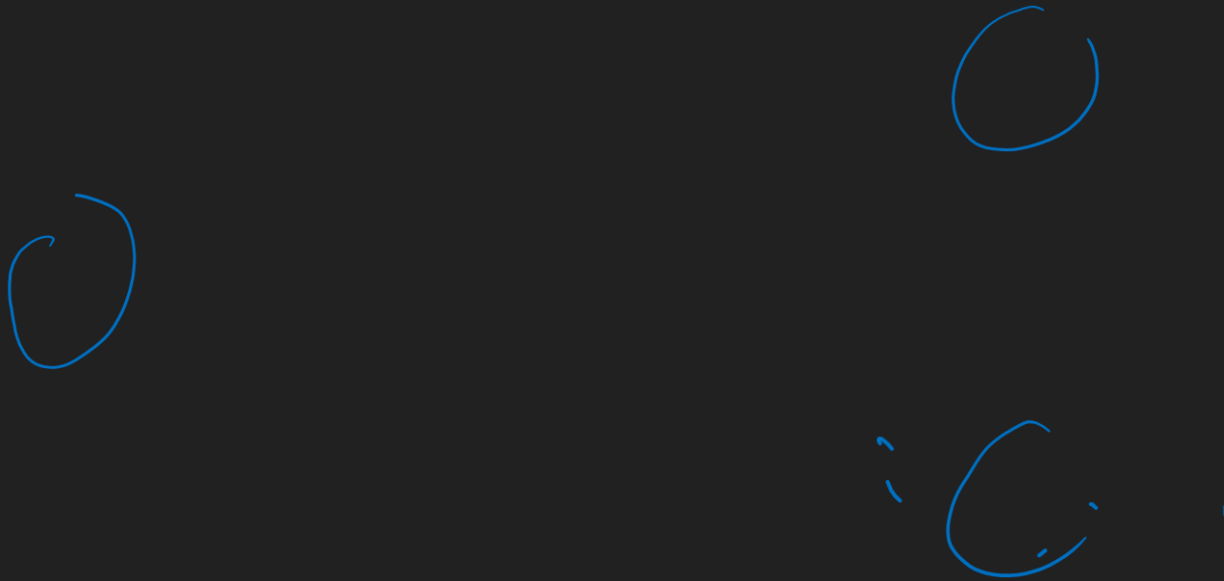
Mouse and human genomes have large blocks of genes in common, and the genes within these blocks are in the same order along the chromosome.

Interpretation:

The different locations of these blocks reflects the rearrangements that have occurred relative to their last common ancestor.



Synteny reflects chromosomal rearrangements that occurred during evolution



Black box above is a video (posted on myCourses), or you can make your own at the link below

<https://wooythan.github.io/syntery-explorer/genome.html?p=EUA&c1=homSap&c2=musMus>